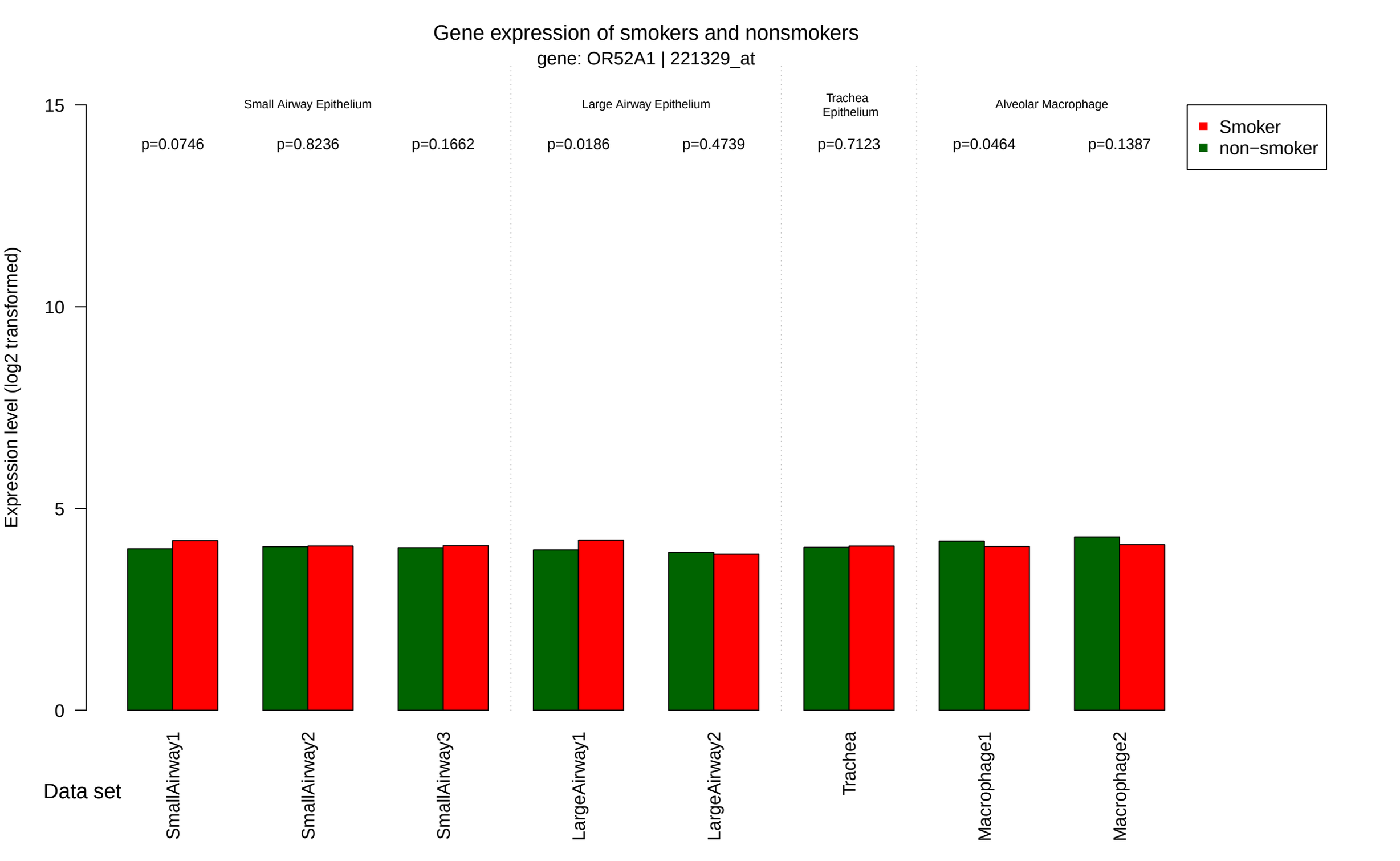
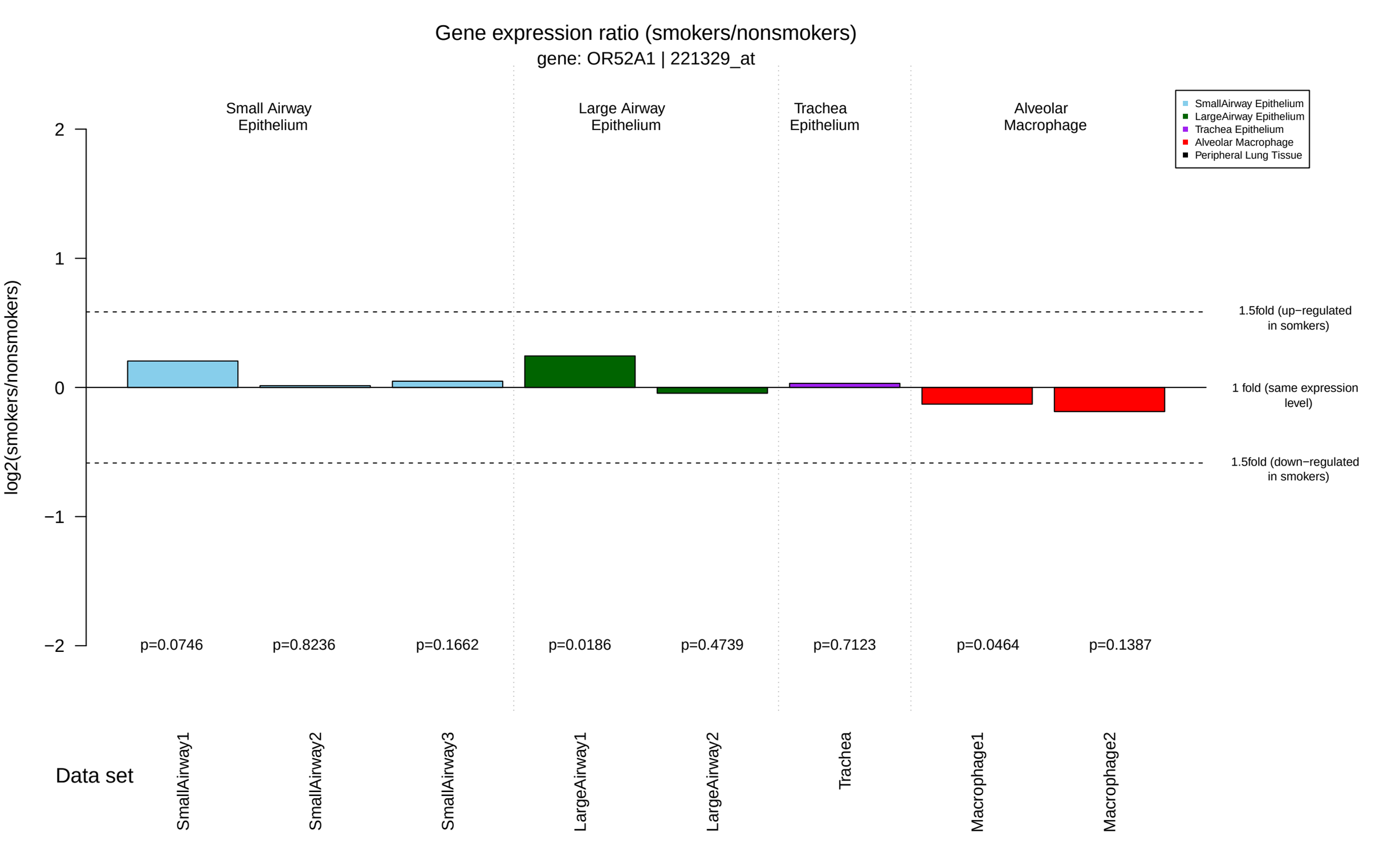
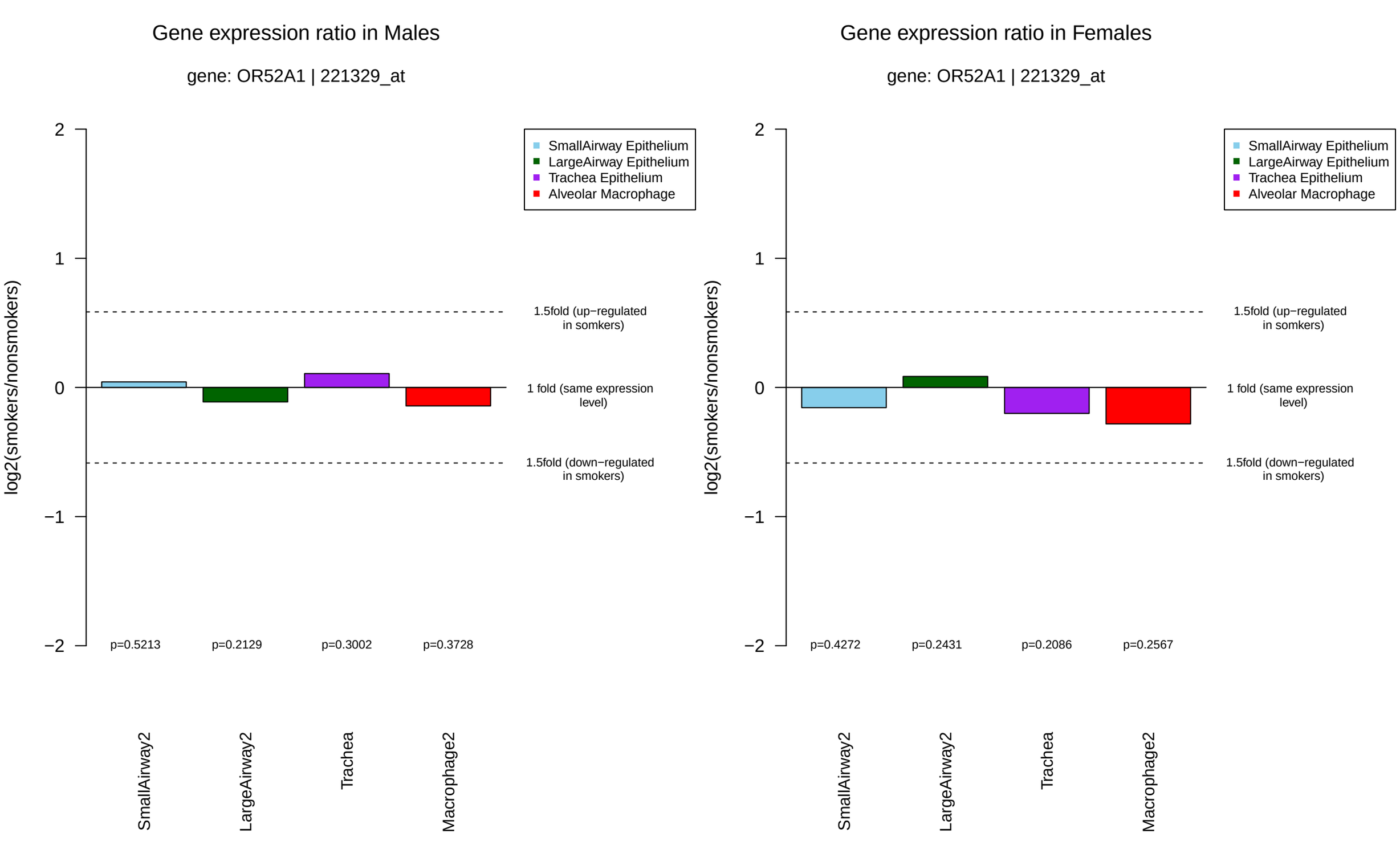
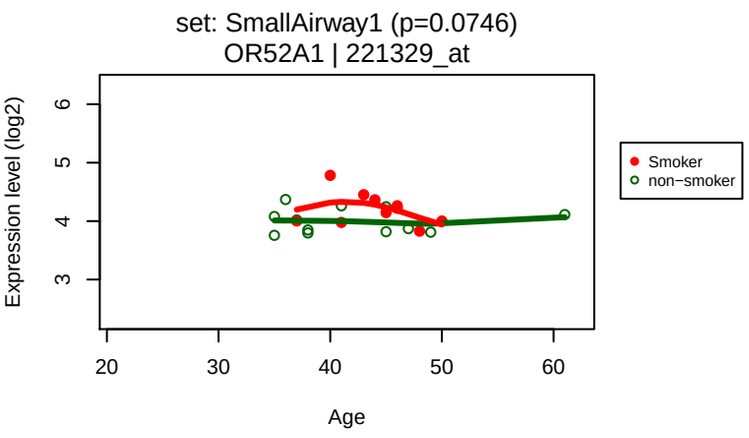
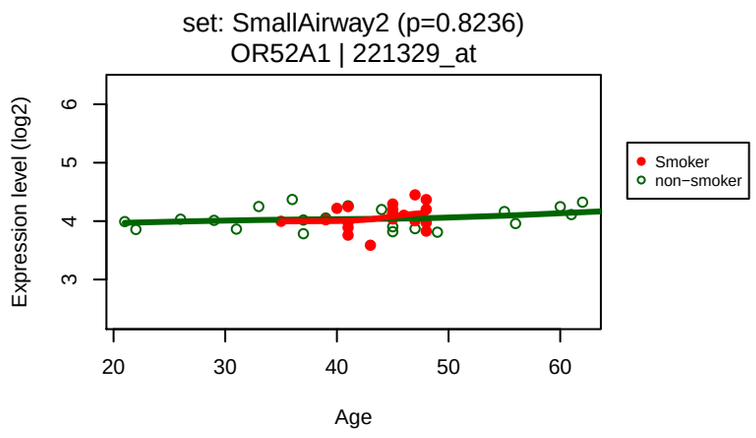
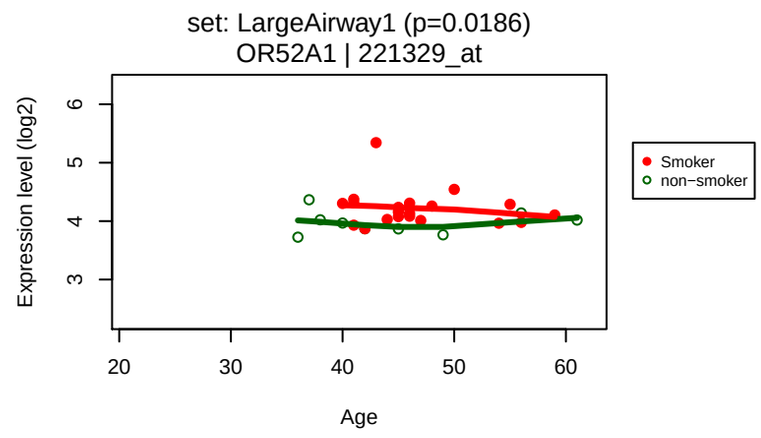

Gene expression of smokers and nonsmokers
gene: OR52A1 | 221329_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0746
p=0.8236
p=0.1662
p=0.0186
p=0.4739
p=0.7123
p=0.0464
p=0.1387
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

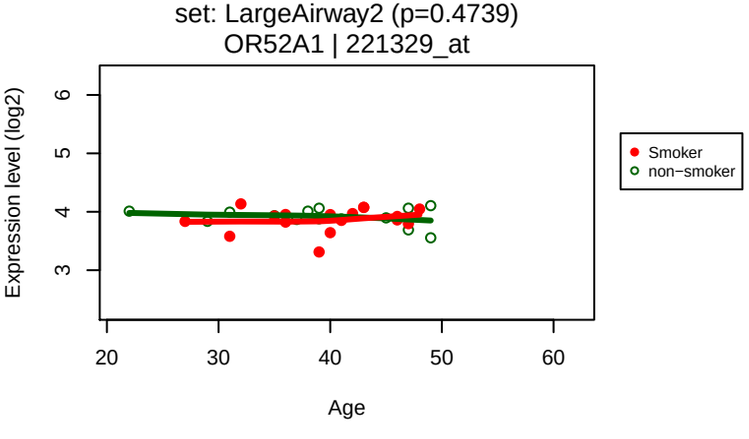

Gene expression ratio (smokers/nonsmokers)
gene: OR52A1 | 221329_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0746
p=0.8236
p=0.1662
p=0.0186
p=0.4739
p=0.7123
p=0.0464
p=0.1387
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

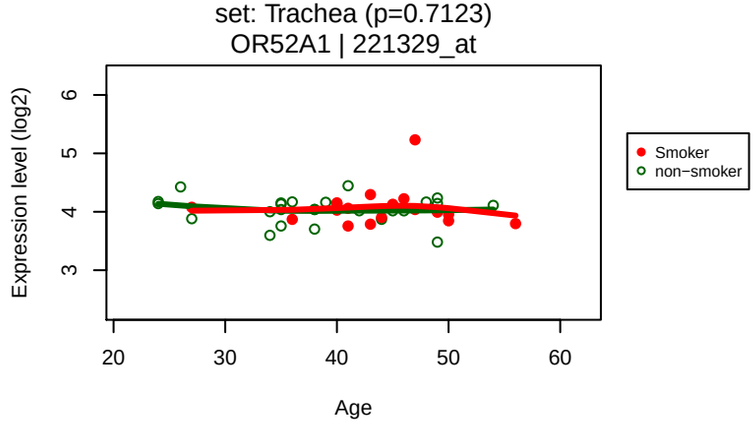

Gene expression ratio in Males
gene: OR52A1 | 221329_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5213
p=0.2129
p=0.3002
p=0.3728
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR52A1 | 221329_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4272
p=0.2431
p=0.2086
p=0.2567
SmallAirway2
LargeAirway2
Trachea
Macrophage2

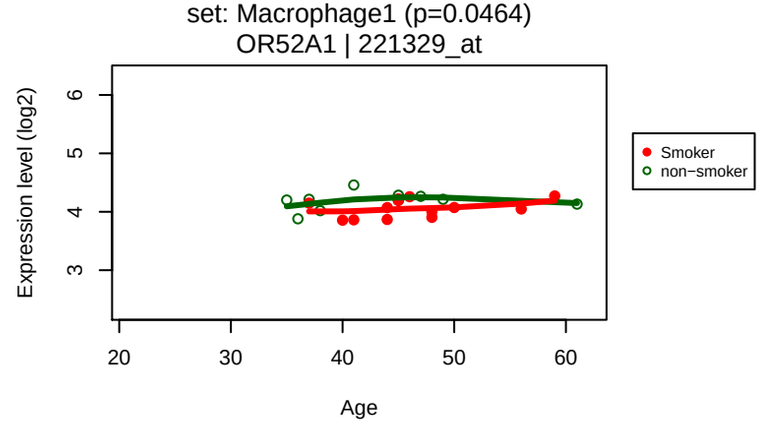

set: SmallAirway1 (p=0.0746)
OR52A1 | 221329_at
Expression level (log2)
Age
Smoker
non−smoker

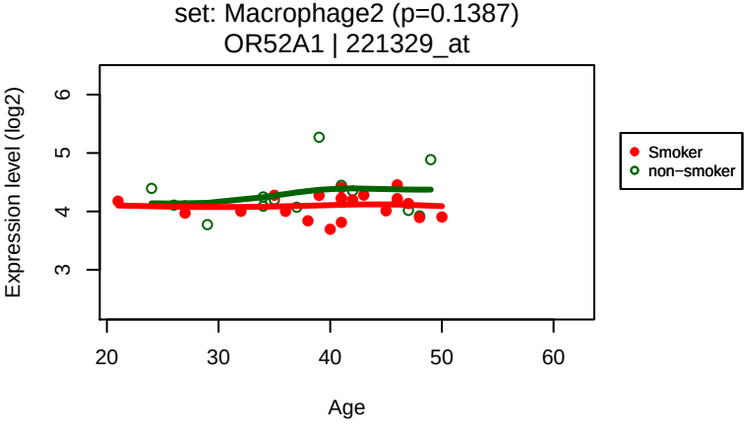

set: SmallAirway2 (p=0.8236)
OR52A1 | 221329_at
Expression level (log2)
Age
Smoker
non−smoker

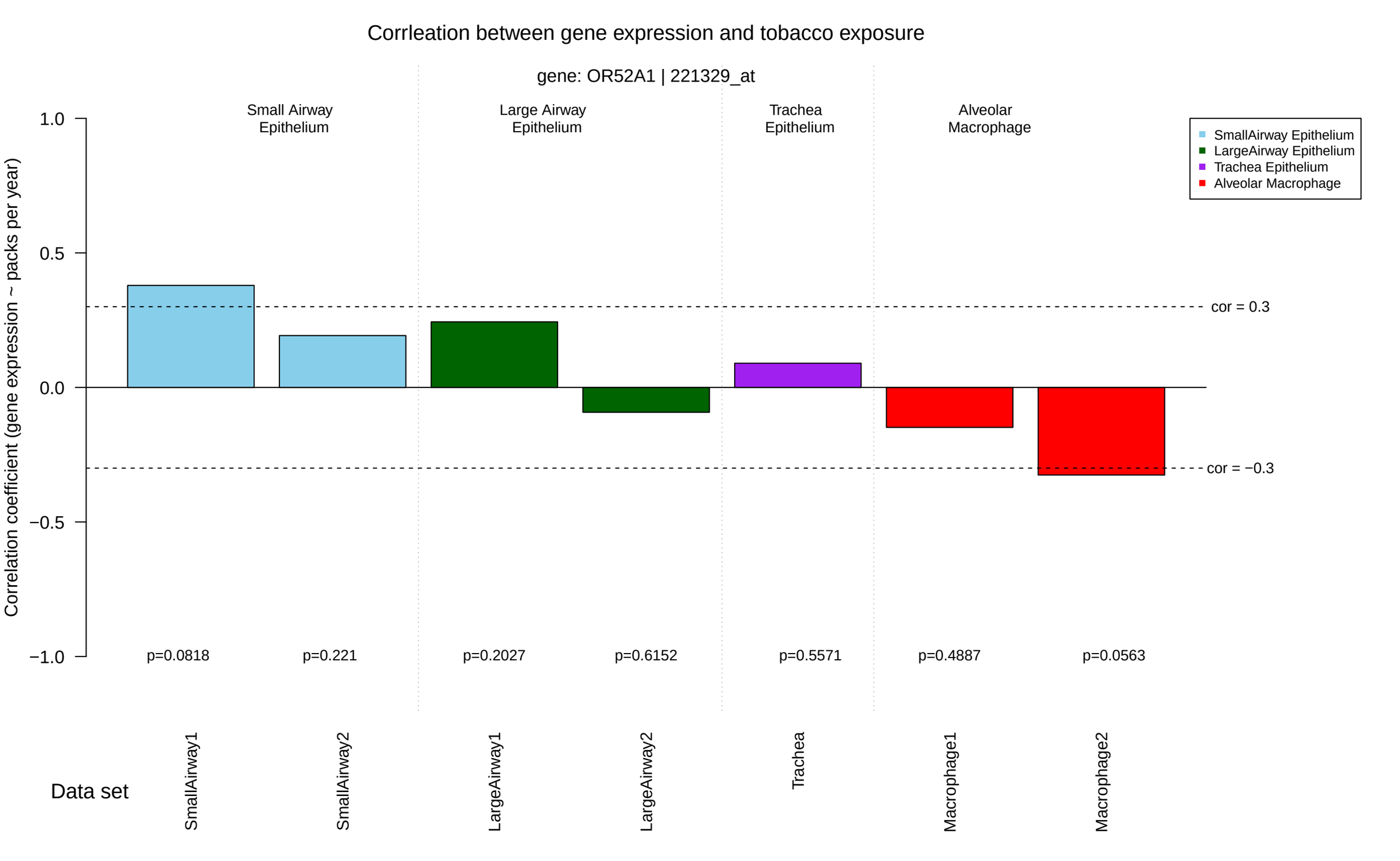

set: LargeAirway1 (p=0.0186)
OR52A1 | 221329_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4739)
OR52A1 | 221329_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7123)
OR52A1 | 221329_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0464)
OR52A1 | 221329_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1387)
OR52A1 | 221329_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR52A1 | 221329_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0818
p=0.221
p=0.2027
p=0.6152
p=0.5571
p=0.4887
p=0.0563
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2